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Abstract

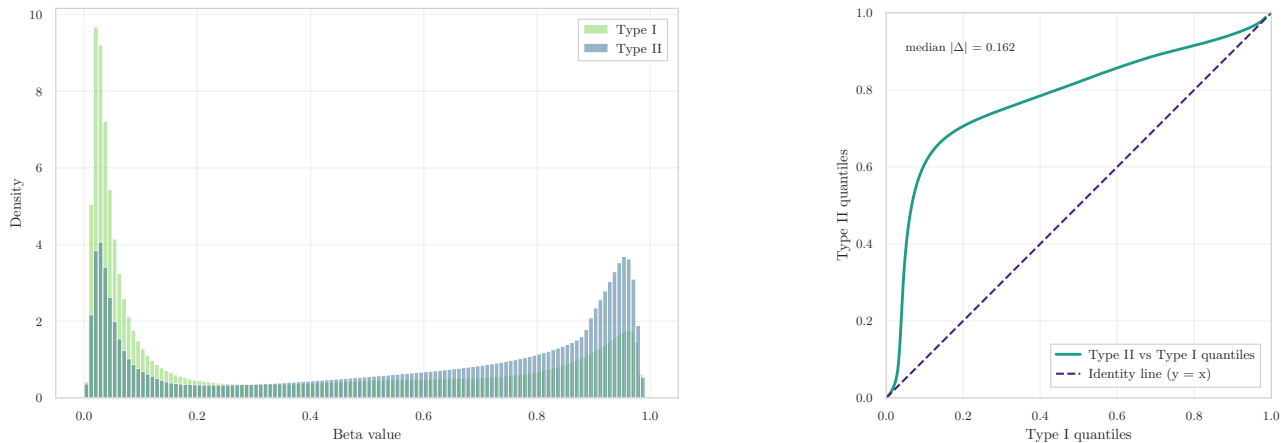
MANCA L'ABSTRACT.

1 Intra-Dataset Pre-processing

Data pre-processing represents a foundational stage in DNA methylation analysis, particularly in cancer epigenomics, where technical artefacts, probe-design biases, and unwanted sources of variation can obscure or distort genuine biological signals. Before performing feature selection, or machine-learning models, it is essential to ensure that the methylation matrices used for statistical modelling are accurate, harmonised across platforms, and free from unreliable probes or batch-associated effects.

In the context of breast cancer, where early epigenetic alterations may be subtle and highly localized, careful pre-processing is crucial to avoid confounding Normal–Adjacent differences with technical noise. Several well-established sources of unreliability—cross-reactive probes, SNP-affected loci, non-CpG targets, Infinium Type I/II probe-design bias, and batch effects—must be addressed through a systematic and reproducible workflow.

The primary goal of this section is therefore to construct, for each dataset, a high-quality and biologically coherent methylation matrix suitable for downstream analyses. To this end, all datasets undergo the same structured pipeline: initial alignment and quality checks; removal of unreliable CpGs through literature-based technical filtering; identification



(a) β -value distribution by Infinium probe design (Type I vs. Type II). The near-overlapping shapes indicate effective correction of probe-type bias. (b) Q-Q plot comparing quantiles of Type II vs. Type I probes. The alignment to the diagonal ($y = x$) confirms balanced signal distributions after normalization.

Figure 1: Diagnostic evaluation of Infinium Type I/II probe bias correction in the GSE69914 dataset. The consistent β -value and quantile distributions demonstrate the effectiveness of the BMIQ normalization previously applied.

of invariant CpGs that carry no discriminative information along the Normal-Adjacent axis; conversion from β -values to M-values for statistical modelling; and correction of batch effects.

These steps ensure methodological consistency across the three cohorts (GSE69914 [1.1], GSE225845 [1.2], GSE287331 [1.3]) and provide a unified, denoised, and biologically meaningful set of features to be used in the subsequent analyses of early epigenetic alterations.

1.1 Dataset GSE69914

Step 0 – Initial Pre-processing The initial pre-processing stage verifies the structural integrity of the dataset. The β -value matrix and its corresponding phenotype table were imported from their Parquet representations. Both tables contained ✓ 397 samples, and the β -matrix included ✓ 485,512 CpGs, confirming full dataset availability at load time.

Sample identifiers in the phenotype table were inspected to ensure uniqueness and correct metadata structure. No duplicated `id_tissue` entries were detected ✓ (0 duplicates), and all required fields (`id_tissue`, `label`) were present ✓ (0 missing fields), indicating that the phenotype annotation is complete and coherent. Alignment between the phenotype table and the β -matrix was assessed by intersecting sample IDs. All samples were shared across matrices, yielding ✓ perfect correspondence (397/397 matched).

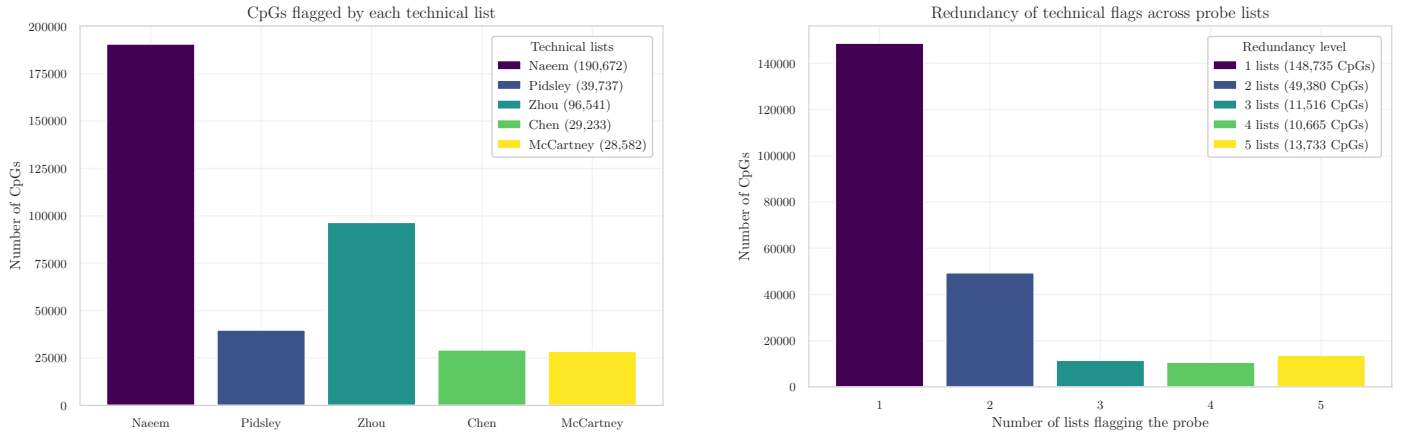
A probe-level and sample-level missingness check was then performed to verify the completeness of the methylation matrix. The entire dataset was found to be fully observed, with ✓ 0 missing CpG values and ✓ 0 missing sample entries. This confirms that no early NaN filtering or imputation is required for this dataset.

Step 1 – Infinium Type I/II Bias Diagnostics Raw methylation intensity data (*IDAT* files) were processed by the original authors using the `minfi` pipeline and BMIQ normalization, as reported in [1]. A well-known characteristic of Illumina Infinium microarrays is the **systematic technical bias between Type I and Type II probes**. This bias has been extensively documented in the literature, most notably by Maksimovic *et al.* Maksimovic2012SWAN, who showed that the two probe chemistries produce inherently different β -value distributions and therefore require appropriate probe-type normalization. Such differences, if uncorrected, can propagate into downstream statistical analyses.

To independently verify that this bias had been successfully mitigated in the present dataset, β -values were stratified by probe design (Type I vs. Type II) using the official Illumina 450K manifest from Bioconductor [2]. Following the diagnostic procedure of Teschendorff *et al.* [3], two complementary assessments were performed:

1. **Density comparison.** As shown in Fig. 1a, the β -value distributions of Type I and Type II probes substantially overlap, indicating the absence of the multi-modal distortion typical of unnormalized data.
2. **Quantile-quantile alignment.** The Q-Q plot in Fig. 1b exhibits quite a diagonal trend for Type II vs. Type I quantiles, which is the expected signature of balanced probe-type scaling after normalization.

The resulting pattern closely matches the characteristic behaviour described by Teschendorff *et al.* [3] and Wang *et al.* [4], while being fully consistent with the probe-type discrepancies originally analysed by Maksimovic *et al.* [5].



(a) CpGs flagged by each technical filtering resource (Naeem, Pidsley, Zhou MASK_*, Chen, McCartney).

(b) Redundancy distribution of probes flagged by 1–5 lists.

Figure 2: Summary of technical filtering for the GSE69914 methylation dataset. **(a)** Intersection between the dataset and each external probe-filtering resource, showing the number of CpGs flagged by Naeem, Pidsley, Zhou, Chen, and McCartney lists. **(b)** Redundancy analysis indicating how many probes were simultaneously flagged by 1 to 5 independent resources, quantifying the concordance among technical annotations.

✓ Together, these diagnostics confirm that the dataset has undergone effective Type I/II probe bias correction.

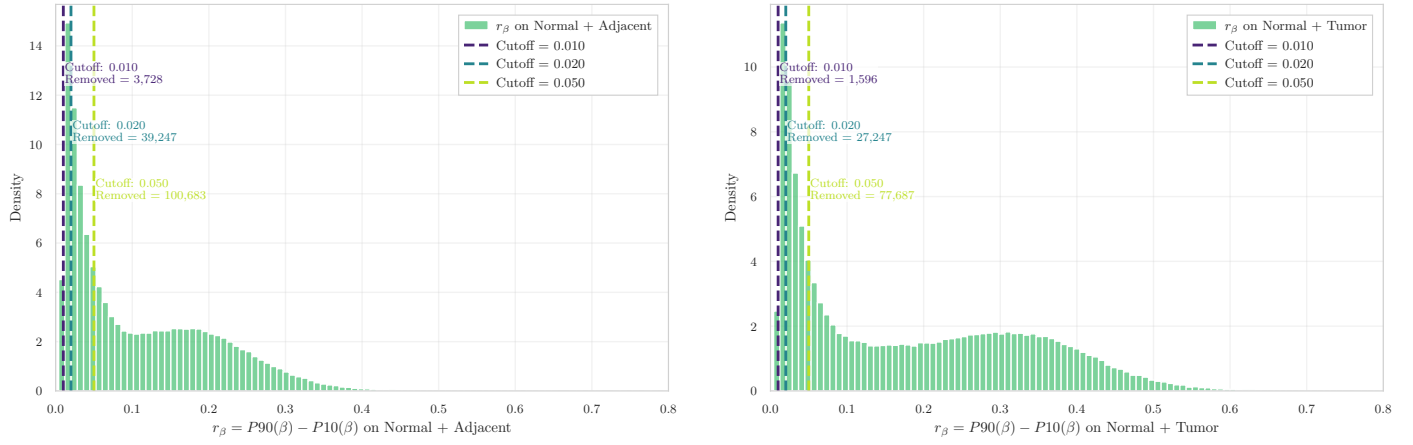
Step 2 – Technical Filtering Technical filtering aims to remove unreliable or biologically confounded probes prior to statistical modelling. This stage reduces noise, improves downstream reproducibility, and ensures that only high-confidence CpG loci are retained for analysis (Fig. 2).

Exclusion of technical probe sets A collection of curated probe resources was used to exclude loci known to exhibit technical artefacts (see Appendix A for details). These include probes affected by common genetic variants at the CpG or single-base extension site [6], [7], probes with demonstrated multi-mapping or off-target hybridisation [8], [9], and probes flagged by the MASK_* reliability annotations [6]. Additional quality control criteria follow the hierarchical filtering strategy of Naeem et al. [10], which targets multi-mapping loci, repeat structures, and disruptive SNPs.

The overlaps between these resources and the β -matrix were substantial (Fig. 2a). A total of ✓ 190,672 probes were flagged by Naeem et al., ✓ 39,737 by Pidsley et al., ✓ 29,233 by Chen et al., ✓ 96,541 by the Zhou MASK_* fields, and ✓ 28,582 by McCartney et al. These lists are known to partially overlap—particularly those focused on multi-mapping and SNP-associated unreliability—and their union resulted in the removal of ✓ 225,426 unique CpGs. After this operation, ✓ 260,086 CpGs remained.

Annotation-based filtering (Illumina manifest) To ensure consistency with the manufacturer’s official annotation, the HumanMethylation450 v1.2 manifest [11] was used to validate probe identity and genomic mapping. Through this annotation, probes lacking valid chromosome information, those with undefined genomic coordinates, and those targeting non-CpG contexts (IDs beginning with “ch”) were identified and removed. This step aligns the working feature set with Illumina’s reference genome definition and follows the recommendations of Zhou et al. [6] and Pidsley et al. [7]. A total of ✓ 875 non-CpG probes were excluded, reducing the feature space to ✓ 259,211 CpGs.

Removal of sex-chromosome probes (X/Y) To avoid sex-related confounding effects and to maintain cross-dataset consistency, all probes located on chromosomes X and Y were removed based on manifest annotations. This choice is consistent with established preprocessing workflows for Illumina 450K data, where probes on sex chromosomes are excluded when male and female samples are mixed, as recommended by Maksimovic et al. [5]. Moreover, sex-chromosome CpGs exhibit substantially different methylation profiles between males and females, and joint normalization of autosomal and sex-chromosome probes has been shown to introduce artificial sex-related bias into thousands of autosomal CpGs [12]. Removing X/Y probes therefore provides a conservative and robust strategy when sex-specific methylation is not under investigation. This filtering step eliminated ✓ 7,728 CpGs, producing a final autosomal working set of ✓ 251,483 CpGs.



(a) r_β distribution on Normal+Adjacent samples, with thresholds at 0.01, 0.02, and 0.05.

(b) r_β distribution on Normal+Tumor samples.

Figure 3: **Variability range analysis (Edgar-style)**. The Normal+Adjacent subset displays a large block of CpGs with extremely low variability ($r_\beta < 0.05$), supporting the use of invariance filtering to reduce biologically uninformative loci.

Technical filtering summary Overall, the initial set of ✓ 485,512 CpGs was reduced to ✓ 251,483 CpGs after applying all technical lists, manifest-based validation, and sex-chromosome exclusion. In total, ✓ 234,029 unique CpGs were removed across all criteria. A per-probe summary file `removed.cpgs.all.filters.summary-gse69914.csv` records, for each excluded probe, which resources contributed to its removal, ensuring full reproducibility of the filtering pipeline.

Step 3 – Invariance Filtering on Normal+Adjacent A stage of feature-level pre-processing was applied to remove CpG loci that exhibit minimal biological variability in **normal breast tissue**. This strategy follows the empirically driven framework proposed by Edgar et al. [13] for detecting non-variable CpGs across tissues using the β -value range statistic

$$r_\beta = P90(\beta) - P10(\beta),$$

computed here separately on **Normal+Adjacent** (N+A) and **Normal+Tumor** (N+T) subsets.

For the N+A group, r_β was computed for all ✓ 251,483 autosomal CpGs. The distribution of variability revealed that a substantial proportion of CpGs display extremely narrow dynamic range in normal breast tissue (Fig. 3). Following Edgar’s recommended threshold of $r_\beta < 0.05$, a total of ✓ 100,683 CpGs (40.0%) were classified as **non-variable** and removed from subsequent modelling, leaving ✓ 150,800 CpGs as the **post-invariance feature set** used to construct the matrix `beta_final_preFS` (✓ 397 samples $\times$ 150,800 CpGs).

A parallel computation on the N+T subset showed reduced invariance: ✓ 77,687 CpGs (30.9%) were non-variable under the same threshold. Comparing the two invariance sets revealed three biologically informative categories:

- **Stable invariants** (✓ 75,029 CpGs): loci consistently non-variable in both N+A and N+T;
- **Breakdown loci** (✓ 25,654 CpGs): CpGs variable only when tumour samples are introduced, indicating possible *tumour-driven epigenetic disruption*;
- **NT-only invariants** (✓ 2,658 CpGs): CpGs non-variable in N+T but variable in normal breast tissue.

To assess whether variability patterns in N+A were consistent with N+T, a scatter comparison between $r_\beta^{(N+A)}$ and $r_\beta^{(NT)}$ was generated (Fig. 4). The two measures exhibited strong concordance (✓ Spearman $\rho = 0.845$), although a clear band of CpGs deviated from the diagonal, corresponding precisely to the **breakdown loci** defined above.

Further annotation-based analysis showed that invariants are not uniformly distributed across genomic elements. Invariant CpGs were enriched in promoter-proximal regions—particularly TSS200 and 1stExon—while gene bodies and 3’UTRs were underrepresented (Fig. 4b). This pattern mirrors the findings of Edgar et al. [13], who reported that invariant CpGs preferentially occupy regulatory elements with constitutively stable methylation profiles.

Invariants in breast tissue were compared with the three tissue invariance sets published by Edgar et al. (blood, buccal, placenta). Their intersection yielded ✓ 42,315 universal CpGs shared across all tissues. Among these, ✓ 30,251 remained invariant in Normal+Adjacent breast, and ✓ 28,351 remained invariant even after including tumour samples. A biologically relevant subset of ✓ 2,010 CpGs belonged to the breakdown category, suggesting possible tissue-specific loss of epigenetic stability in tumourigenesis.

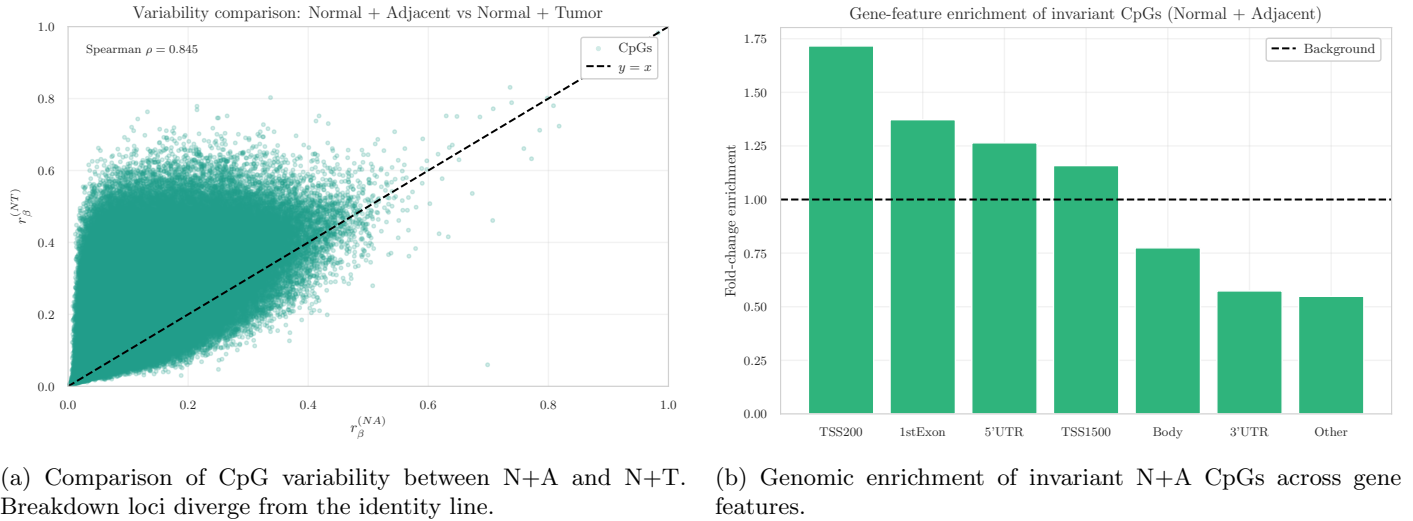


Figure 4: **Variability concordance and genomic context of invariant CpGs.** Stable loci dominate the diagonal, while breakdown loci reflect tumour-induced methylation variability. Promoter regions show the strongest enrichment of invariant CpGs.

Step 4 – β to M Conversion Illumina Infinium methylation arrays measure methylated ($y^{(M)}$) and unmethylated ($y^{(U)}$) signal intensities per CpG. The processed dataset provides only β -values,

$$\beta = \frac{y^{(M)}}{y^{(M)} + y^{(U)}},$$

a bounded proportion in $[0, 1]$ characterised by strong mean–variance dependence. Following the foundational analysis of Du *et al.* [14], all β -values were transformed into M -values using the logit relation

$$M = \log_2 \left(\frac{\beta + \varepsilon}{1 - \beta + \varepsilon} \right), \quad \varepsilon = 10^{-6},$$

which yields an approximately homoscedastic scale more suitable for linear modelling, empirical Bayes moderation, ComBat batch correction, and regression-based inference.

Importantly, the adoption of M -values is not only theoretically motivated but also widely implemented in practice. Several high-impact studies routinely perform statistical modelling in M -space while retaining β -values solely for biological interpretation and effect-size visualisation. Examples include the differential methylation analysis of purified human blood cell types by Reinius *et al.* [15], the methodological framework underlying the `minfi` pipeline described by Triche *et al.* [16], and oncological applications such as the tumour hypoxia study by Thienpont *et al.* [17], where all β -values were explicitly transformed to M -values prior to statistical testing. These examples illustrate that M -values have become a de facto standard for inferential analyses in Illumina 450K/EPIC workflows.

The conversion was applied to the complete post-filtering matrix (✓ 150,800 CpGs across ✓ 397 samples). A manual inspection of transformed values confirmed correct numerical behaviour across the full CpG panel, including loci with extreme β close to 0 or 1.

Distributional consequences of the transformation. Fig. 6 illustrates in detail how the $\beta \rightarrow M$ mapping reshapes the empirical distribution of methylation. Normal and Adjacent tissues display the expected U-shaped profiles in β -space, reflecting biological accumulation near fully unmethylated or fully methylated states. After logit transformation, the corresponding M -value distributions become substantially more symmetric and comparable across groups, indicating that the transformation regularises the scale without distorting biological structure.

The mean–SD relationships further highlight this effect. In β -space, variance is strongly mean-dependent and inflates near the boundaries, whereas M -values exhibit markedly flatter SD profiles. This variance-stabilising behaviour is consistent with the theoretical findings of Du *et al.* [14] and with the practical motivations underlying the adoption of M -values in modern methylation studies.

Step 5 – Batch Effect Correction

Step 6 – Final Outputs

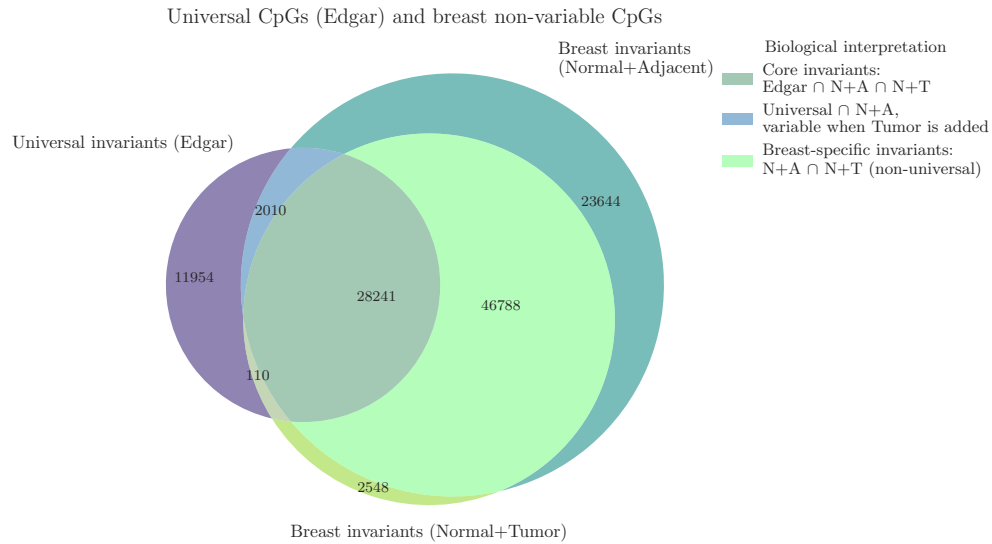


Figure 5: **Intersection between universal invariant CpGs (Edgar) and breast invariants.** Core invariants ($\text{Edgar} \cap \text{N+A} \cap \text{N+T}$) represent strongly conserved loci, whereas “breakdown loci” highlight CpGs losing stability specifically in tumour tissue.

1.2 Dataset GSE225845

Step 0 — Initial Pre-processing

Ingestion and Alignment

Coarse NaN Assessment

Infinium Type I/II Bias Diagnostics

Step 1 — Technical Filtering

Cross-reactive Probes

EPIC Unreliable Probes

SNP-affected Probes

Non-CpG Probes

Sex-chromosome Probes

Technical Filtering Summary

Step 2 — β -to- M Conversion

Definition of β_{clean}

Logit Transformation

Usage of β and M -values

Step 3 — Batch Effect Correction

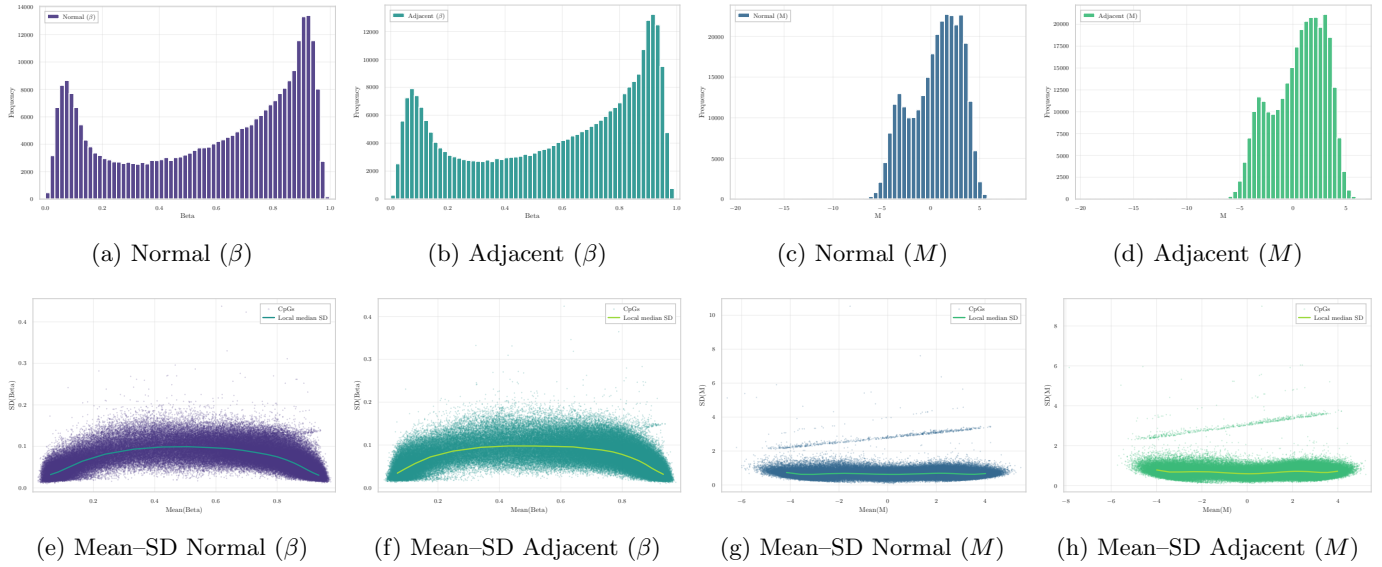


Figure 6: **Distributional comparison between β and M quantifications in Normal and Normal-adjacent tissues.** Top row: β exhibits a bounded, U-shaped distribution, whereas M shows an approximately symmetric profile. Bottom row: The substantial mean-dependent variance in β is dramatically reduced after transformation, confirming that M -values provide a more homoscedastic scale for statistical modeling. All panels computed on the working set of ✓ 150,800 CpGs after complete technical and invariance filtering.

Pre-batch Diagnostics

Batch Model Definition

ComBat Application

Post-batch Verification

Step 4 — Invariance Filtering on Normal+Adjacent

NA Subset Selection

Variability Metrics

Threshold Selection

Final CpG Mask

Step 5 — Final Outputs

Final Matrices

CpG Tracking

Mini-report Summary (optional)

1.3 Dataset GSE287331

Step 0 — Initial Pre-processing

Ingestion and Alignment

Coarse NaN Assessment

Infinium Type I/II Bias Diagnostics

Step 1 — Technical Filtering

Cross-reactive Probes

EPIC Unreliable Probes

SNP-affected Probes

Non-CpG Probes

Sex-chromosome Probes

Technical Filtering Summary

Step 2 — β -to- M Conversion

Definition of β_{clean}

Logit Transformation

Usage of β and M-values

Step 3 — Batch Effect Correction

Pre-batch Diagnostics

Batch Model

ComBat Application

Post-batch Verification

Step 4 — Invariance Filtering on Normal+Adjacent

NA Subset Selection

Variability Metrics

Threshold Selection

Creation of Final Mask

Step 5 — Final Outputs

Final Matrices

CpG Tracking

Mini-report Summary (optional)

2 Inter-Dataset Analysis

3 Data Validation and Integrity Check

Missing Value Analysis. Next, I performed a comprehensive check for missing values (NaN), as these can severely impact model performance and must be addressed before training.

- No missing entries (NaN) were detected across any CpG probe. ✓ Total NaN count: 0 — Overall missing rate: 0%.
- The methylation matrix is therefore **complete**, requiring **no filtering or imputation** procedures. ✓ Dataset confirmed fully complete.
- For future datasets:
 - If the overall missing rate is $< 1\%$, imputation may be considered as an optional step.
 - If probe missingness exceeds 5% or sample missingness exceeds 10%, the affected entities should be discarded, following standard preprocessing practices [6].

This validation confirms the dataset is structurally sound, numerically consistent, and complete, enabling unbiased downstream variance modeling, differential methylation testing, and batch correction without any further cleaning.

A Filtering lists

This appendix details the major technical filtering lists and annotations applied in Illumina 450K and EPIC methylation arrays to remove unreliable or ambiguous CpG probes.

Table 1: Technical categories covered by each filtering resource.

Category	Naeem A.1	Chen A.2	Pidsley A.3	Zhou A.4	McCartney A.5
Cross-hybridisation / multi-mapping	Yes (Steps 1–2)	Yes	Yes	MASK_mapping	Table 2+3
SNP at CpG / extension base	Yes (Step 5)	–	Yes	MASK_snp5 , MASK_extBase	–
Nearby SNP tolerated	Conditional (Step 6)	–	–	–	–
INDEL / structural variant	Yes (Step 3)	–	–	–	–
Non-CpG probes	–	–	Yes	MASK_nonCG	Table 3

A.1 Reducing the risk of false discovery enabling identification of biologically significant genome-wide methylation status using the humanmethylation450 array

Naeem *et al.* [10] proposed a structured filtering framework to reduce false discoveries by sequentially discarding probes based on hybridisation specificity and genomic integrity (Figure 7). The main exclusion steps are:

- Multi-mapping or cross-hybridising probes $\Rightarrow$ *discard*.
- Probes overlapping repetitive elements (LINE/SINE/ALU) $\Rightarrow$ *discard*.
- Probes targeting regions with INDELs $\Rightarrow$ *discard*.
- Probes overlapping SNPs:
 - SNP at CpG or extension site $\Rightarrow$ *discard*.
 - SNP near target but not interfering in bisulfite space $\Rightarrow$ *keep*.

The resulting “discard” set therefore removes probes affected by cross-reactivity, structural polymorphisms (INDELs), and SNPs that alter CpG interrogation.

A.2 Discovery of cross-reactive probes and polymorphic cpGs in the illumina infinium humanmethylation450 microarray

Chen *et al.* [8] empirically identified probes in the 450K array that exhibit off-target hybridisation or overlap with common SNPs. For this project, only the **cross-reactive list** is available and used. This list enumerates ~ 29 k multi-mapping probes whose methylation signal cannot be uniquely attributed to one genomic locus.

A.3 Critical evaluation of the Illumina MethylationEPIC BeadChip microarray for whole-genome DNA methylation profiling

Pidsley *et al.* [7] provide a platform-level assessment of the EPIC array, with explicit *annotated probe lists* that flag probes whose methylation signal may be confounded by design- or genome-related artefacts. The focus is on probe categories and positions where technical bias arises, rather than on a universal, prescriptive blacklist. In particular, they catalogue:

- **Cross-hybridising CpG-targeting probes:** CpG probes showing sequence homology (off-target matches) to additional genomic loci, yielding non-unique hybridisation and potentially inflated or ambiguous β signals.
- **Cross-hybridising non-CpG-targeting probes:** off-target issues among CNG/non-CpG probes; these are typically excluded in CpG-centric analyses due to limited interpretability and higher risk of artefacts.
- **Probes overlapping common genetic variation:** annotation of variants from population data at three critical positions:
 - *At the interrogated CpG* (polymorphic CpG) — directly affects the presence of the CpG dinucleotide and the measured methylation state;

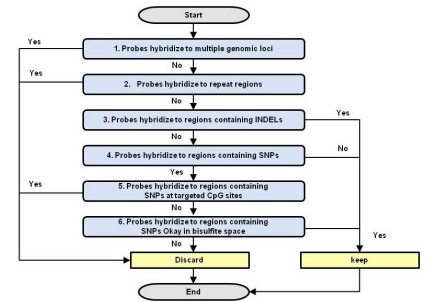


Figure 7: Workflow for determining affected Probes.

- *At the single-base extension (SBE) site* (Type I) — perturbs extension and dye chemistry, biasing intensity ratios;
- *Within the probe body* — can reduce binding affinity or alter hybridisation kinetics, especially for common variants.

A.4 Comprehensive characterization, annotation and innovative use of infinium dna methylation beadchip probes

Zhou *et al.* [6] released a unified probe annotation for both 450K and EPIC arrays with multiple binary mask columns (MASK_*). Each mask flags probes to be removed due to specific technical artifacts.

- MASK.mapping: probes with low or inconsistent mapping quality (non-unique alignment or presence of INDELs);
- MASK.typeINextBaseSwitch: Type-I probes carrying a SNP in the extension base that causes a color-channel switch (CCS probes);
- MASK.extBase: probes whose extension base is inconsistent with the expected color channel or CpG context;
- MASK.sub30.copy: probes with non-unique 30-bp 3' subsequences (potential cross-hybridisation);
- MASK.snp5.common: probes overlapping a SNP within ± 5 bp of the interrogated CpG (even with global MAF $\geq 1\%$);
- MASK.snp5.GMAF1p: probes overlapping SNPs with global MAF $\geq 1\%$;
- MASK.general: recommended composite mask integrating mapping, SNP, and cross-reactivity filters for general use;
- MASK.rmsk15: probes overlapping RepeatMasker regions (not recommended for exclusion in standard workflows).

This resource provides a programmatic and reproducible way to perform fine-grained probe masking.

A.5 Identification of polymorphic and off-target probe binding sites on the Illumina Infinium MethylationEPIC BeadChip

McCartney *et al.* [9] assessed probe design artifacts and published supplementary tables that include:

- Cross-hybridising CpG-targeting probes;
- Cross-hybridising non-CpG-targeting probes.

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